

Printing months in the x axis with pyplot

Asked 3 years, 2 months ago Modified 3 years, 2 months ago Viewed 4k times



Data I'm working with: <https://drive.google.com/file/d/1xb7icmocz-SD2Rkq4ykTZowxW0uFFhBl/view?usp=sharing>

1

Hey everyone,



I am a bit stuck with editing a plot. Basically, I would like my x value to display the months in the year, but it doesn't seem to work because of the data type (?). Do you have any idea how I could get my plot to have months in the x axis?



If you need more context about the data, please let me know!!!



Thank you!

Here's my code for the plot and the initial data modifications:

```
import matplotlib.pyplot as plt
import mplleaflet
import pandas as pd
import matplotlib.dates as mdates
from matplotlib.dates import DateFormatter
import numpy as np

df =
pd.read_csv("data/C2A2_data/BinnedCsvs_d400/fb441e62df2d58994928907a91895ec62c2c42ef

df['degrees']=df['Data_Value']/10

df['Date'] = pd.to_datetime(df['Date'])

df2 = df[df['Date']<'2015-01-01']

df3 = df[df['Date']>='2015-01-01']

max_temp = df2.groupby([(df2.Date.dt.month),(df2.Date.dt.day))]['degrees'].max()
min_temp = df2.groupby([(df2.Date.dt.month),(df2.Date.dt.day))]['degrees'].min()

max_temp2 = df3.groupby([(df3.Date.dt.month),(df3.Date.dt.day))]['degrees'].max()
min_temp2 = df3.groupby([(df3.Date.dt.month),(df3.Date.dt.day))]['degrees'].min()

max_temp.plot(x='Date', y='degrees', kind='line')
min_temp.plot(x='Date',y='degrees', kind='line')

plt.fill_between(range(len(min_temp)),min_temp, max_temp, color='C0', alpha=0.2)

ax = plt.gca()

ax.set(xlabel="Date",
      ylabel="Temperature",
      title="Extreme Weather in 2015")

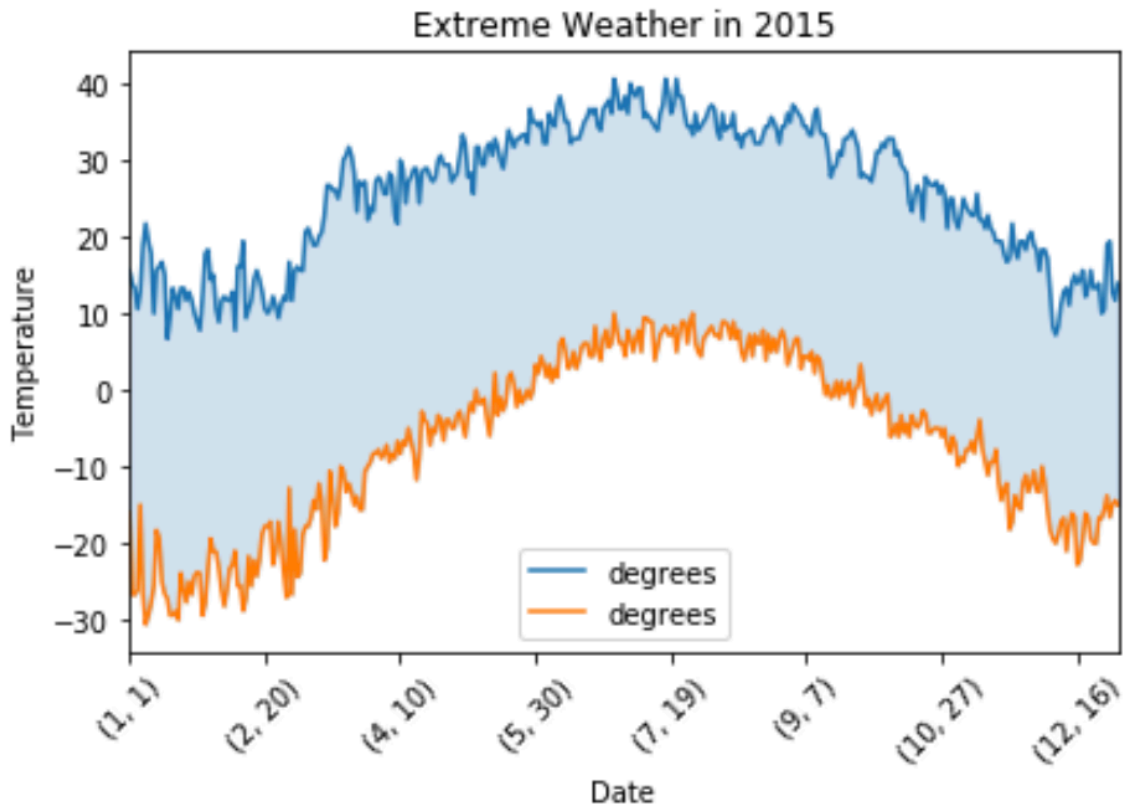
plt.legend()
plt.tight_layout()

x = plt.gca().xaxis

for item in x.get_ticklabels():
    item.set_rotation(45)

plt.show()
```

Plot I'm getting:



[python](#) [pandas](#) [date](#) [plot](#) [series](#) [Edit tags](#)

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edited Jun 27, 2021 at 19:19

asked Jun 27, 2021 at 19:17



Henry Ecker ♦
35.3k 19 44 62



pegasus123
99 1 7

1 ▲ What do you want the labels to look like? 01-01, Jan-01, etc? – Henry Ecker ♦ Jun 27, 2021 at 19:23

1 ▲ @HenryEcker Jan, Feb and so on! Do you know how to? – pegasus123 Jun 27, 2021 at 19:40

1 Answer

Sorted by: [Reset to default](#) [Date modified \(newest first\)](#)



Option 1 (Most Similar Approach)

1

Change the index based on month abbreviations using [Index.map](#) and [calendar](#)



This is just for df2 :

```
import calendar
```

```
import matplotlib.pyplot as plt
import pandas as pd
```



```
df = pd.read_csv(...)
df['degrees'] = df['Data_Value'] / 10
df['Date'] = pd.to_datetime(df['Date'])
df2 = df[df['Date'] < '2015-01-01']
```

```
max_temp = df2.groupby([df2.Date.dt.month, df2.Date.dt.day])['degrees'].max()
min_temp = df2.groupby([df2.Date.dt.month, df2.Date.dt.day])['degrees'].min()
```

```
# Update the index to be the desired display format for x-axis
max_temp.index = max_temp.index.map(lambda x: f'{calendar.month_abbr[x[0]]}')
min_temp.index = min_temp.index.map(lambda x: f'{calendar.month_abbr[x[0]]}')

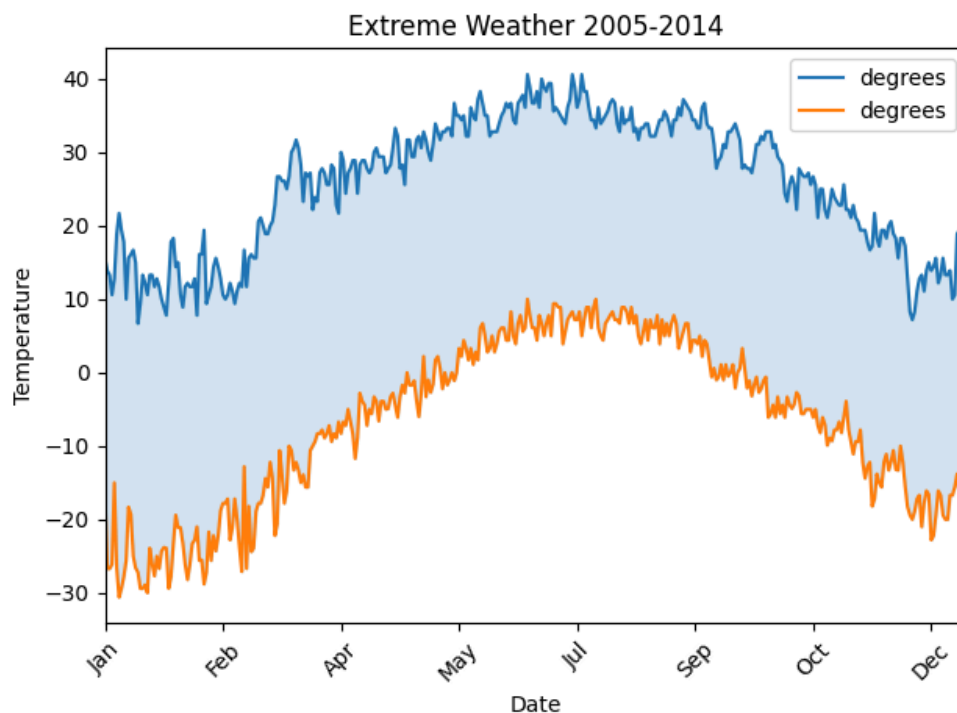
max_temp.plot(x='Date', y='degrees', kind='line')
min_temp.plot(x='Date', y='degrees', kind='line')

plt.fill_between(range(len(min_temp)), min_temp, max_temp,
                 color='C0', alpha=0.2)

ax = plt.gca()
ax.set(xlabel="Date", ylabel="Temperature", title="Extreme Weather 2005-2014")

x = plt.gca().xaxis
for item in x.get_ticklabels():
    item.set_rotation(45)

plt.margins(x=0)
plt.legend()
plt.tight_layout()
plt.show()
```



As an aside: the title "Extreme Weather in 2015" is incorrect because this data includes all years *before* 2015. This is "Extreme Weather 2005-2014"

The year range can be checked with `min` and `max` as well:

```
print(df2.Date.dt.year.min(), '-', df2.Date.dt.year.max())
# 2005 - 2014
```

The title could be programmatically generated with:

```
title=f"Extreme Weather {df2.Date.dt.year.min()}-{df2.Date.dt.year.max()}"
```

Option 2 (Simplifying groupby step)

Simplify the code using [groupby aggregate](#) to create a single DataFrame then convert the index in the same way as above:

```

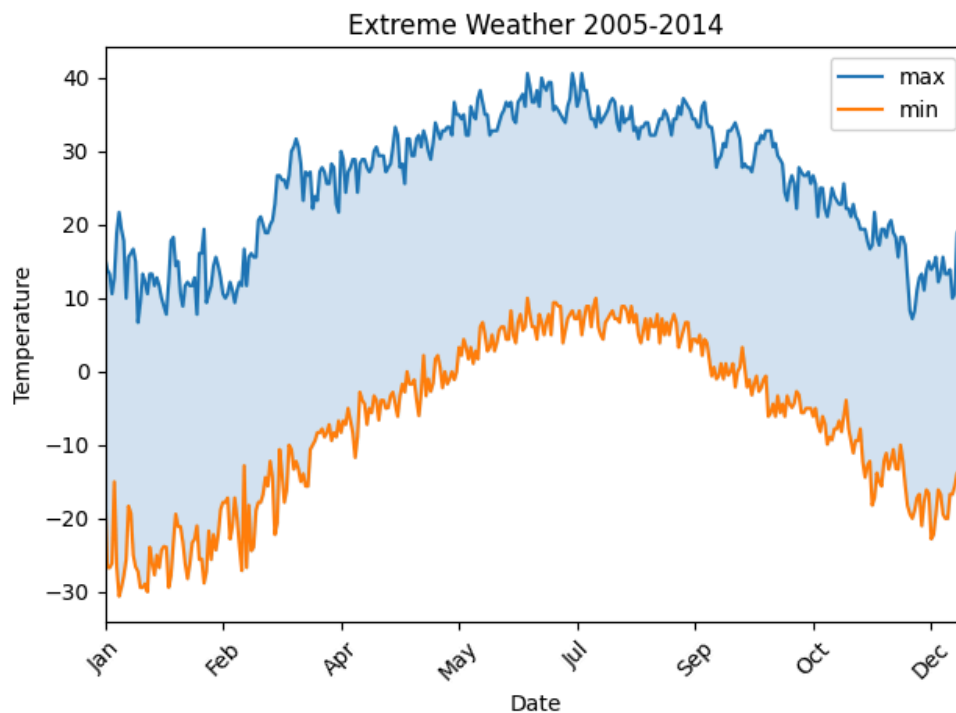
import calendar

import matplotlib.pyplot as plt
import pandas as pd

df = pd.read_csv("...")
df['degrees'] = df['Data_Value'] / 10
df['Date'] = pd.to_datetime(df['Date'])
df2 = df[df['Date'] < '2015-01-01']
# Get Max and Min Degrees in Single Groupby
df2_temp = (
    df2.groupby([df2.Date.dt.month, df2.Date.dt.day])['degrees']
    .agg(['max', 'min'])
)
# Convert Index to whatever display format is desired:
df2_temp.index = df2_temp.index.map(lambda x: f'{calendar.month_abbr[x[0]]}')

# Plot
ax = df2_temp.plot(
    kind='line', rot=45,
    xlabel="Date", ylabel="Temperature",
    title=f"Extreme Weather {df2.Date.dt.year.min()}-{df2.Date.dt.year.max()}"
)
# Fill between
plt.fill_between(range(len(df2_temp)), df2_temp['min'], df2_temp['max'],
                color='C0', alpha=0.2)
plt.margins(x=0)
plt.tight_layout()
plt.show()

```



Option 3 (Best overall functionality)

Convert the index to a datetime using [pd.to_datetime](#). Choose any leap year to uniform the data (it must be a leap year so Feb-29 does not raise an error). Then set the `set_major_formatter` using the format string `%b` to use the month abbreviation:

```

import matplotlib.dates as mdates
import matplotlib.pyplot as plt
import pandas as pd

df = pd.read_csv("...")
df['degrees'] = df['Data_Value'] / 10
df['Date'] = pd.to_datetime(df['Date'])
df2 = df[df['Date'] < '2015-01-01']

```

```

# Get Max and Min Degrees in Single Groupby
df2_temp = (
    df2.groupby([df2.Date.dt.month, df2.Date.dt.day])['degrees']
    .agg(['max', 'min'])
)

# Convert to DateTime of Same Year
# (Must be a leap year so Feb-29 doesn't raise an error)
df2_temp.index = pd.to_datetime(
    '2000-' + df2_temp.index.map(lambda s: '-'.join(map(str, s)))
)

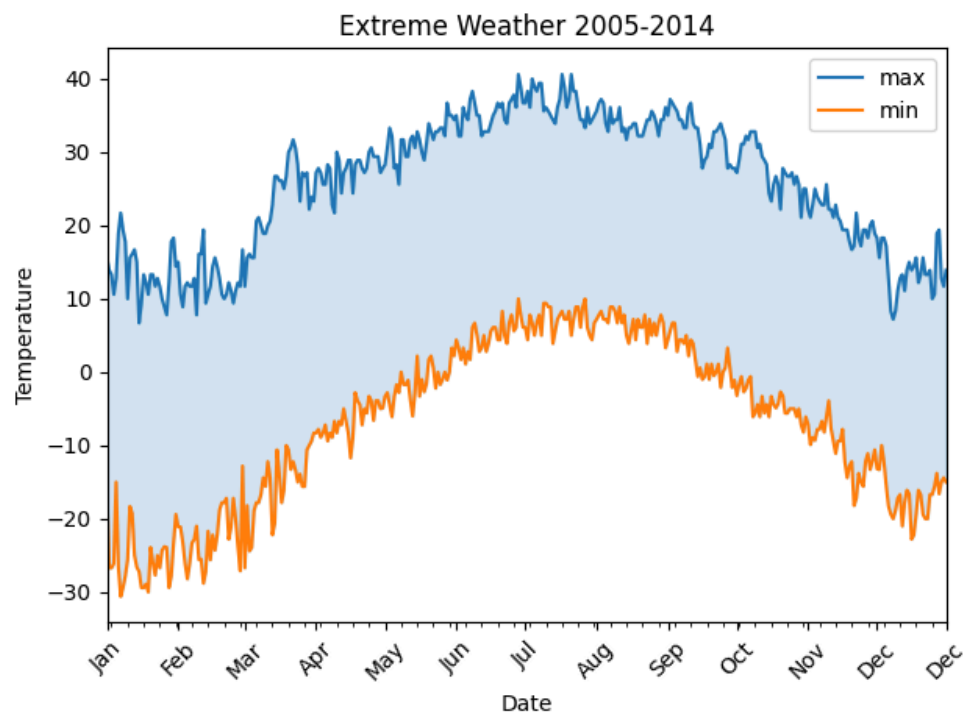
# Plot
ax = df2_temp.plot(
    kind='line', rot=45,
    xlabel="Date", ylabel="Temperature",
    title=f"Extreme Weather {df2.Date.dt.year.min()}-{df2.Date.dt.year.max()}"
)

# Fill between
plt.fill_between(df2_temp.index, df2_temp['min'], df2_temp['max'],
    color='C0', alpha=0.2)

# Set xaxis formatter to month abbr with the %b format string
ax.xaxis.set_major_formatter(mdates.DateFormatter('%b'))
plt.tight_layout()
plt.show()

```

The benefit of this approach is that the index is a datetime and therefore will format better than the string representations of options 1 and 2.



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edited Jun 27, 2021 at 21:17

answered Jun 27, 2021 at 19:25



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